Measuring Underwater Image Quality with Depthwise-separable Convolutions and Global-sparse Attention

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***Abstract*—Underwater Image Quality Assessment (UIQA) poses specific challenges due to blur, dispersion of light, and color garbling caused by water turbulence. Images captured by Autonomous Underwater Vehicles (AVS) correlate poorly along their RGB channels, as blue and green channel components dominate. We propose a novel lightweight no-reference (NR) UIQA architecture based on depthwise-separable convolutions and global sparse attention that surpasses the existing deep learning architectures in terms of performance as well as parameter count and Multiply-Accumulate operations (MACs). Proposed model gives 3.64% increase in SRCC and 6.11% increase in PLCC in comparison to state-of-the-art for UEQAB dataset. Our work also illustrates the effectiveness of depthwise separable convolutions for underwater image quality assessment. Our empirical analysis confirms less redundancy among the channels of a feature map when depthwise separable convolutions are used compared to standard convolutions by increasing representational efficiency.**

***Index Terms*—Underwater Image Quality Assessment (UIQA), Global Sparse Attention, Depthwise-Separable Convolution, PLCC Loss, Rank Loss**

I. Introduction

Underwater imagery technology has applications in diverse domains like marine biology, oceanography, environmental monitoring, and underwater inspection and maintenance. In these areas, accurate and reliable image data is critical for making informed decisions. Underwater environments present unique challenges for imaging systems due to factors such as low light, limited visibility, and water turbulence, which can affect the quality and clarity of the images [14]. As a result, it is important to have a systematic approach to grading the quality of underwater images to ensure their suitableness for purposes like oceanography and marine inspection.

Underwater image quality assessment (UIQA) is a vital task that ensures the accuracy, reliability, and validity of underwater image data for a spectrum of scientific, commercial, and recreational activities. Subjective assessment of image quality is the most widely used method to grade images, as in most cases, human observers have most trusted visual systems. However, this method takes time and effort. Nonetheless, objective assessment of image quality involves using mathematical models that mimic the human evaluation of an input image. This method saves time and resources. Reduced-reference (RR), full-reference (FR), and no-reference (NR) are three objective image quality assessment (IQA) varieties. In addition, quality assessment can help identify problems and limitations with underwater imaging systems and provide guidance for technological improvements and advancements.

This work proposes a computationally efficient deep learning network using depthwise-separable convolution and global-sparse attention for no reference UIQA. The proposed network's performance is evaluated using UEQAB [8] dataset. We give an empirical analysis of inter-channel correlation of feature map from intermediate layers of proposed model to assess the effectiveness depthwise-separable convolution for UIQA.

II. Related Work

The proposed deep learning based UIQA model addresses the research gaps identified during the study of available hand-crafted features based UIQA metrics. Existing deep learning architectures for IQA are covered in this section as there is limited work done using deep learning for UIQA. Also, deep networks that used depthwise-separable convolutions for increasing performance are studied.

A. Underwater IQA Metrics

To grade underwater images, some manually crafted features have been added to raw scene measures, such as NIQE [11] and codebook-based image quality (CBIQ) [26]. Jiang *et al.* [4] proposed underwater image quality measure by extracting quality-acquainted attributes from luminance and chrominance details. With a color assessment in the frequency and spatial domains, Yang *et al.* [24] suggest a traditional formulation-based UIQA measure. Although underwater image quality can be measured using an assortment of contrast, saturation, and chroma, this measure is called Underwater Color Image Quality Evaluation (UCIQE). Underwater image quality measure (UIQM) is a quality assessing metric developed by Panetta *et al.* [13] and uses a linear blend of three quality measurements to assess contrast, sharpness, and colorfulness. Moreover, Wang *et al.* [18] introduced a UIQA measure known as CCF, that draws inspiration from imaging techniques and to indicate the color distortion due to the fog generated by rearward dispersal, the blurring induced by onward dispersion and absorption; this method integrates fog density index, contrast index, and colorfulness index. Mengyao *et al.* [8] proposed the first deep learning-based algorithm for UIQA.

B. Deep Learning based IQA architectures

Blind image quality assessment (BIQA) [17] from DeepRN utilizes a convolutional neural network (CNN). Yan *et al.* [22] introduce a CNN with two streams that accept a gradient map and image as distinct inputs to handle the issue of quality attribute deduction from one input and permit the accession of distinguishable decks of details from the input. A large artificial distortion-based dataset with comprehended image grading is used to train a dual (Siamese) network, known as RankIQA [10], that automatically measures the level of ranked distortion without manual interference. Lin and Wang [9] proposed NR-IQA that uses a distorted image to render a fictitious reference image in order to equalize the absence of an authentic reference image employing Generative Adversarial Networks (GAN). You and Korhonen [27] utilizes techniques for learning positional embedding adaptively to assess images of varying resolutions with the help of exploration based on vision transformer in image quality measurement (TRIQ). Yang *et al.* [25] presented a Multi-dimension Attention Network for NR-IQA (MANIQA) to enhance scores on grimaces caused by GANs. Further, image quality transformer (IQT) [1] introduced by Cheon *et al.*, which effectively utilizes a transformer architecture for FR-IQA. Moreover, Su *et al.*'s [16] proposed approach introduces a hyper network (HyperIQA) with fused multi-scale features to designate image quality in real-world scenarios by dividing the quality prediction process into two stages. Though these deep learning (DL) models proved to be effective for the quality assessment of natural images, there is a need for effective DL architectures that considers the properties of underwater images.

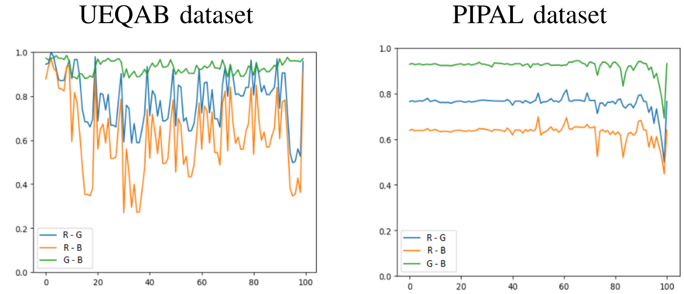
This paper contributes the following to the UIQA domain:

1) A novel NR-UIQA deep learning model with global sparse attention and depthwise-separable convolutions.
2) Empirically analysis of the increase in representational efficiency when depthwise-separable convolutions are used instead of standard convolutions for UIQA.

The rest of the paper is arranged as follows. Section 3 discusses the proposed architecture with the idea behind using depthwise-separable convolutions in deep IQA methods. Empirical analysis and experimental results to exhibit the success behind depthwise-separable convolutions are illustrated in section 4. Further, the conclusion is given in Section 5.

III. Proposed Methodology

The RGB color channels of natural images are highly correlated, and the same kernels can be used across the different channels to extract meaningful information. At the same time, the RGB color channels of underwater images are not highly correlated with each other. To validate this argument, Fig. 1 shows the correlation between the RGB channels of natural images (from PIPAL dataset [5]) and underwater images (from UEQAB dataset [8]) and it confirms



(a) (b)

Fig. 1: Analyzing R-G, R-B, and G-B channel correlation for 100 randomly selected images of each dataset. Unlike natural images in the PIPAL dataset, the G-B plot is independent of R-G and R-B for underwater images in UEQAB dataset. This confirms Green and Blue channels are highly correlated and dominate in the case of underwater images.

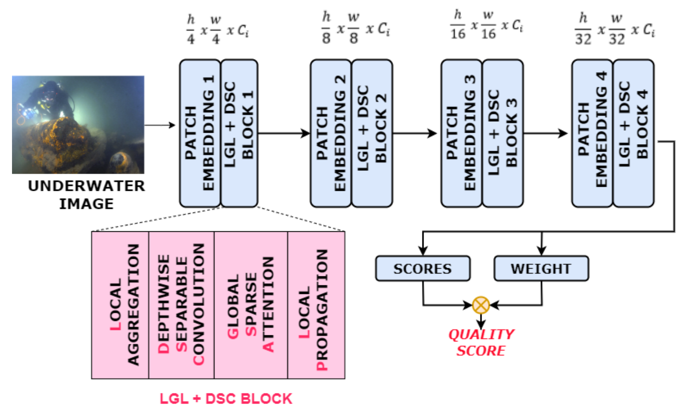


Fig. 2: The overall architecture of the proposed algorithm.

green and blue component dominates compared to red in case of underwater water images. With this view, using the same kernel for the different channels of underwater is not highly efficient, as these channels are containing different information. Hence, to comprehend this trait of underwater images, we propose the use of depthwise separable convolution rather than standard convolution. Depthwise separable convolution employs separate kernels for depthwise convolution and pointwise convolution. This gives an edge while operating on spatial locations which are highly correlated and channels that are less correlated with each other. We propose a lightweight pyramidal transformer architecture that attempts to decrease the feature map's size and enrich the information present across the channels.

A. Proposed Architecture

The proposed architecture for quality assessment of underwater images consists of the following components: Patch Embedding, Local-Global-Local + Depthwise-Separable Convolution block, and dual branch prediction head. The detailed architecture is shown in Fig. 2.

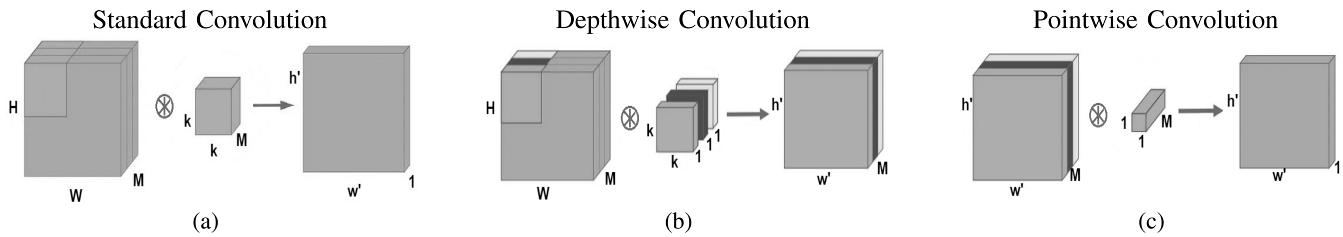


Fig. 3: Illustration of Standard Convolution and Depthwise Separable Convolution. Pointwise Convolution preceded by Depthwise Convolution constitutes Depthwise Separable Convolution. Here, ⊛ symbol represents the convolution operation.

- *Patch Embedding Block*: These are formed by convolution layer and layer normalization. It acts as a downsampling unit.
- *Local-Global-Local (LGL) information exchange + Depthwise-Separable Convolution Block*: LGL is adopted from EdgeViTs [12], and a depthwise separable convolution layer is included to use information present across the channels. This consists of components, including local aggregation, global sparse attention, depthwise-separable convolutions, and local propagation.
  - *Local Aggregation*: Information is accumulated in the target token from neighboring tokens of a local window k x k.
  - *Global Sparse Attention*: A subset of tokens from evenly distributed spatial space is selected instead of using all spatial tickets in the window. Keles *et al.* [7] indicated that the complexity of self-attention in a transformer is quadratic; thus, global sparse attention proposed by Pan *et al.* [12] is used. A single token from the window of size r x r is chosen; r is the sub-sample rate for the stage.
  - *Depthwise Separable Convolution*: In the above steps, only spatial content is taken into account. A layer of depthwise-separable convolution is availed to leverage the information present across different channels.
  - *Local Propagation*: A transposed convolution distributes the collected information among neighboring tokens.
- *Prediction Head*: It is a dual-branch component consisting of parallel weight and score branches proposed by Yang *et al.* [25]. The weight branch consists of a linear layer, ReLU, linear layer, and sigmoid in the same order. This is in parallel to the scoring branch consisting of identical blocks except sigmoid replaced with ReLU.
- The deep-learning architecture described above is trained with a loss function, and this loss function is a combination of correlation and root mean square error loss. Wu *et al.* [20] illustrates the correlation loss has shown better performance as compared to the other loss functions. With this view, we propose to integrate correlation loss with the proposed algorithm. The loss function used in the proposed algorithm is as follows:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{PLCC}} + \mathcal{L}_{\text{SRCC}} + \mathcal{RMSE} \tag{1}$$

$$\mathcal{L}_{\text{PLCC}} = 1 - \text{PLCC}(y, \hat{y}),$$

$$\mathcal{L}_{\text{SRCC}} = 1 - \text{SRCC}(y, \hat{y}),$$

$$\mathcal{RMSE} = \sqrt{\sum_{i=1}^{L} \frac{(\hat{y}_i - y_i)^2}{L}}$$

where y and $\hat{y}$ are the ground truth and predicted perceptual quality scores.

The weight decay and optimizer used for the training of the above-described architecture is 1×10^{-5} for learning rate 6×10^{-4} and Adam optimizer, respectively. The batch size was 16, and the sub-sample rate across four stages is 4,2,2,1 respectively. The five randomly cropped input image patches' quality scores are averaged to give the final predicted score.

B. Empirical Analysis to understand the effectiveness of Depthwise Separable Convolution for UIQA

As discussed earlier, the correlation between the different RGB channels of underwater images is very limited. To understand the effect of this limited correlation on the different convolution layers of the deep-learning architectures, empirical analysis is presented. Depthwise separable convolutions have demonstrated impressive results in image classification models. They have enabled the creation of models that outperform previous counterparts, such as the Xception architecture [2], while also significantly reducing the number of parameters needed to achieve a certain level of performance as seen in the MobileNets [3] family. Additionally, machine translation architectures [6] utilize convolutional sequence-to-sequence networks and have yielded favorable outcomes.

The term "depthwise separable convolution" refers to a type of convolutional operation that operates on both spatial dimensions and the depth dimension (i.e., the number of channels). This technique involves breaking a kernel into two kernels, which perform the depthwise convolution followed by pointwise convolution.

These are represented mathematically by Kaiser *et al.* [6] as follows:

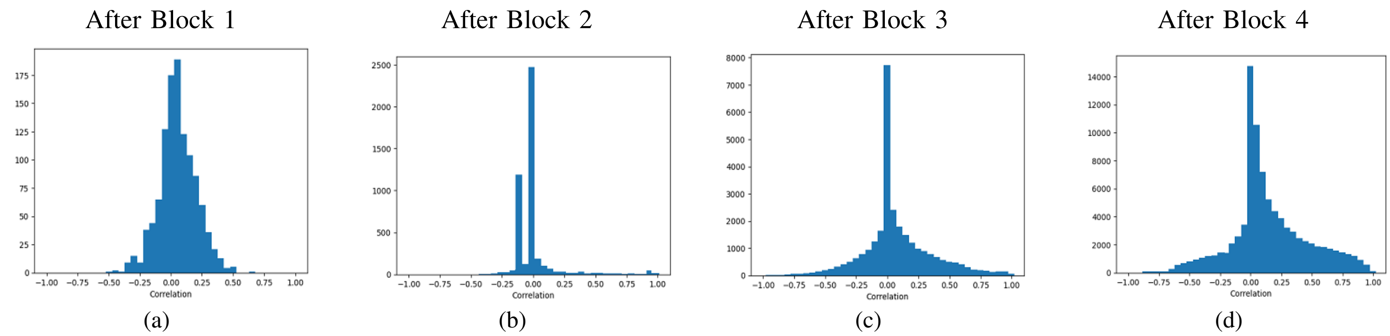


Fig. 4: Inter-channel correlation distribution for intermediate layers of the proposed model with Depthwise Separable Convolutions on underwater image. As the underwater image moves from blocks in (a) to (d), the maximum density is close to 0, which implies less redundancy in channels (i.e., better representation) and, thus, boosts the performance.

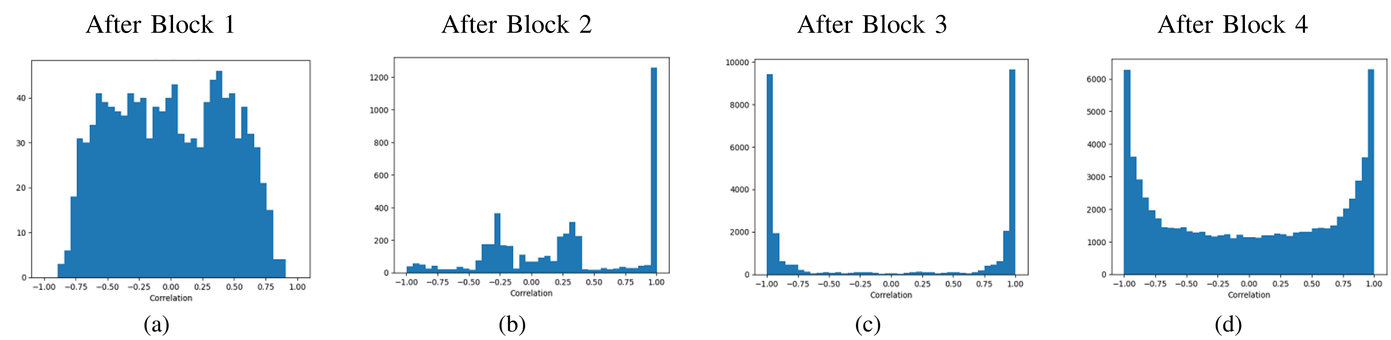


Fig. 5: Inter-channel correlation distribution for intermediate layers of the proposed model with Standard Convolutions on the underwater image. As the underwater image traverses from blocks in (a) to (d), the distribution spreads over the range of -1 to 1 which indicates redundant information is introduced, degrading the performance.

$$\mathrm{SC}(\mathcal{Q},\mathcal{I})_{(i,j)} = \sum_{c,l,b}^{C,L,B} \mathcal{Q}_{(c,l,b)} \cdot \mathcal{I}_{(i+c,j+l,b)} \tag{2}$$

$$\mathrm{DC}(\mathcal{Q},\mathcal{I})_{(i,j)} = \sum_{c,l}^{C,L} \mathcal{Q}_{(c,l)} \odot \mathcal{I}_{(i+c,j+l)} \tag{3}$$

$$\mathrm{PC}(\mathcal{Q},\mathcal{I})_{(i,j)} = \sum_{b}^{B} \mathcal{Q}_{b} \cdot \mathcal{I}_{(i,j,b)} \tag{4}$$

$$\mathrm{DSC}\left(\mathcal{Q}_p, \mathcal{Q}_d, \mathcal{I}\right)_{(i,j)} = \mathrm{PC}_{\ (i,j)}\left(\mathcal{Q}_p, \mathrm{DC}_{\ (i,j)}\left(\mathcal{Q}_d, \mathcal{I}\right)\right) \tag{5}$$

Here SC , PC , DC, DSC refer to standard convolution, pointwise convolution, depthwise convolution, and depthwise-separable convolutions, respectively. These are also illustrated in Fig. 3. $\mathcal{Q}$ denotes kernel, c, l, b denote dimension of kernel, $\mathcal{I}$ denotes input image, $\odot$ denotes element-wise multiplication.

In this work, we consider three other IQA models - MANIQA [25], IQT [1], HyperIQA [16] along with a proposed model for assessing the impact of using depthwise-separable convolution instead of standard convolution for UIQA. The reason behind the effectiveness of depthwise separable convolution is illustrated with respect to the correlation between channels in feature maps of the DL architecture's intermediate layers for UIQA. The feature maps (dimension: height (H) x width (W) x channels (N)) from intermediate layers are extracted. Then, we compute Structural Similarity Index Measure (SSIM) [19] among channels by taking two out of N channels at a time and repeating it for all the channels in the feature map. SSIM close to zero means channels are independent and carry different information, and close to 1 or -1 means they are either positively or negatively correlated [15]. Unlike MAE, MSE, or cosine similarity, SSIM is not a distance metric; thus, it is most suitable for representing inter-channel correlation. Further, the distribution of SSIM scores across channel pairs of each intermediate layer shows a remarkable difference between the distributions obtained using standard convolutions and those obtained with depthwise separable convolutions.

Fig. 4 and 5 show the distribution of feature map correlation collected from various intermediate blocks for both depthwise-separable convolution and standard convolution, respectively. As we traverse down the network layers, distribution shifts closer to -1 or 1 in the case of standard convolution and closer to 0 when depthwise separable convolution layers are used. In these histograms of SSIM score showing channel correlation is on the x-axis, and the y-axis depicts the density of the channel pairs of the feature map having that amount of correlation.

No correlation among channels implies less redundancy of information constrained among the channels of the feature maps. Thus, we are representing more information in channels when depthwise separable convolutions are used. We conclude that in the case of underwater images, depthwise-separable convolutions lead to low inter-channel dependencies, eventually carrying more information and increasing the representational efficiency.

IV. Result Analysis

A. UEQAB Dataset

In order to check the performance of the proposed algorithm, the UEQAB dataset [8] is used. The UEQAB dataset comprises 800 underwater images, sized 256×256, each of these images is enhanced using nine different underwater enhancement algorithms and it yields to a total of 8000 underwater images. These 8000 images are available, along with the mean opinion score (MOS) associated with each image. In their work [8], the authors define UIQA as state-of-the-art with metrics PLCC equal to 0.85 and SRCC equal to 0.85. This database contains a rich pool of underwater images generated using the latest underwater enhancement algorithms, one of the largest known datasets for UIQA. In order to conduct efficient analysis, the UEQAB dataset is fragmented into 3 parts training, testing, and validation data in the ratio of 8:1:1.

B. Performance Analysis

TABLE I: Performance comparison of proposed architecture with other metrics for UEQAB dataset.

Methods		SRCC	PLCC
Handcrafted features based UIQA metrics	UICM [13]	-0.149	-0.272
	UISM [13]	0.261	0.174
	UIConM [13]	0.268	0.151
	UCIQE [23]	0.412	0.359
Deep Learning based quality assessment architectures	***Proposed***	***0.881***	***0.902***
	MANIQA [25]	0.834	0.862
	HyperIQA [16]	0.831	0.862
	UIQAN [8]	0.85	0.85
	IQT [1]	0.741	0.806
	BFEN [21]	0.81	0.81
	NUIQ [21]	0.65	0.65

TABLE II: Parameter count and MACs of proposed UIQA model with other architectures of comparable performance.

Architecture	#Params (in Millions)	#MACS (in Billions)
Proposed	***14***	***2.3***
MANIQA [25]	116	103
HyperIQA [16]	107	8.6
IQT [1]	1700	2.5

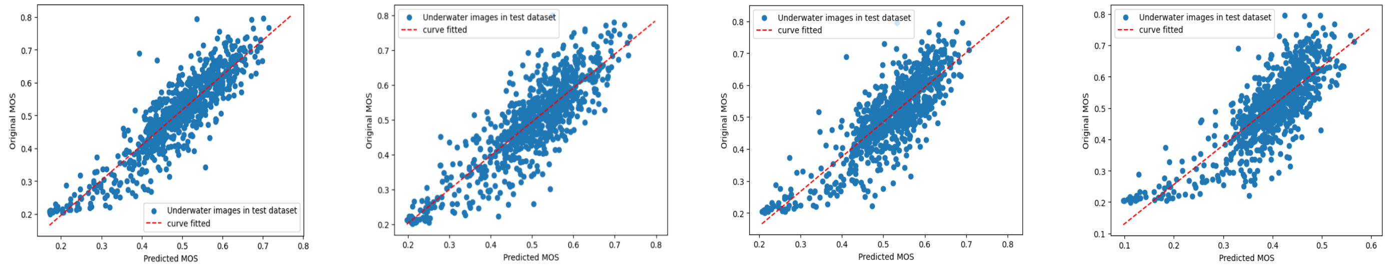
Table I represents the comparison (in terms of SRCC and PLCC) of the proposed algorithm with the 4 handcrafted feature-based UIQA metrics - UICM [13], UISM [13], UIConM [13], UCIQE [23]; 6 deep learning based quality assessment architecture (retrained on UEQAB dataset) MANIQA [25], HyperIQA [16], UIQAN [8], IQT [1], BFEN [21], and NUIQ [21]. From this Table, it can be observed that the proposed algorithm performs better than the existing algorithms. Also, in order to show that the inclusion of Depthwise Separable Convolution layers improves the performance of the proposed algorithm, an ablation study is shown in Table III. For this purpose, in the architectures of MANIQA [25], IQT [1], and HyperIQA [16], the standard convolution layers are replaced with the Depthwise Separable Convolution layers. From this Table, it can be observed that replacing the standard convolution layers with Depthwise Separable Convolution layers significantly improves the performance of these algorithms. In order to show that using the global sparse attention can significantly reduce the number of trainable parameters and Multiply-Accumulate operations (MACs), analytical study is presented in Table II. From this Table, it can be observed that the proposed algorithm requires training a much lesser number of parameters as compared to the MANIQA [25], IQT [1], and HyperIQA [16]. Fig. 6 represents the consistency between ground truth mean opinion scores and the predicted quality scores using the proposed algorithm and the MANIQA [25], HyperIQA [16], and IQT [1]. These results clearly validate the replacement of standard convolution with depthwise separable convolution for the purpose of quality assessment of underwater images.

TABLE III: Percentage increase in performance of deep learning architectures for UIQA when Depthwise-separable convolution is employed in place of standard convolution

Deep Learning IQA Architecture	SRCC	PLCC
Proposed - Standard Convolution	0.854	0.869
Proposed - Depthwise Separable Convolution	0.881	0.902
Percentage Increase	3.161%	3.797%
MANIQA - Original	0.834	0.862
MANIQA - Depthwise Separable Convolution	0.874	0.891
Percentage Increase	4.796%	3.361%
IQT - Original	0.741	0.806
IQT - Depthwise Separable Convolution	0.756	0.813
Percentage Increase	2.02%	0.868%
HyperIQA - Original	0.831	0.862
HyperIQA - Depthwise Separable Convolution	0.844	0.875
Percentage Increase	1.564%	1.508%

V. Conclusion

Our work illustrates a novel lightweight model that leverages depthwise-separable convolutions and global sparse attention for underwater image quality assessment tasks, leading to SRCC equal to 0.881 and PLCC equal to 0.902 . The proposed network leads to an increase of 3.64% in SRCC and 6.11% in PLCC compared to the state-of-the-art. In this work, we have also empirically analyzed how depthwise separable convolutions are more suitable for underwater images as compared to the standard convolution layers. The proposed architecture also performs better in terms of parameter count and MACs compared to other DL architectures for UIQA tasks. This work can be extended to underwater video quality



(a) Proposed Architecture (b) MANIQA [25] (c) HyperIQA [16] (d) IQT [1]

Fig. 6: Scatter plot of predicted MOS by various UIQA methods and original MOS of UEQAB test set.

assessment tasks and can be integrated with underwater image enhancement methods.

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